Phase 2g: One-Loop Soliton–Defect Transmission Phase, Closed-Form Defect Stability Spectrum, and KP-Triangle Fusion for the Good Boussinesq Equation

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Abstract

We compute the one-loop ($O(\hbar^0)$) soliton–defect transmission phase for the good Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ on the Corrigan–Zambon (CZ) integrable type-I defect (Route E, step 2 of `report.pdf`). Every nontrivial claim is labelled **(a)** proved in the literature, **(b)** conjectured/heuristic in the literature, **(c)** derived and verified in this project, or **(d)** a guess.

Our main results **(c)** are:

1. **Exact time-dependent phonon scattering on N-soliton + defect backgrounds.** Extending the CZ 1- and 2-soliton defect solutions to 3 solitons and linearising in an oscillatory wave $E_3 = \varepsilon\, \mathrm{e}^{\mathrm{i}(qx-\omega(q)t)}$ ($\omega(q) = |q|\sqrt{1+q^2}$) produces an exact time-dependent phonon scattering eigenmode satisfying all four linearised CZ sewing conditions at $x = 0$ for all $t \in \mathbb{R}$, with zero reflection ($R(q) = 0$), zero asymptotic evanescent tails, and factorised transmission amplitude $T_{\mathrm{tot}}(q) = \big(\prod_{j=1}^{N} A(j, \mathrm{ph}(q))\big) z_{\mathrm{ph}}(q; q_D, \eta)$.
2. **KP-vertex factorisation and 2-root dipole decomposition.** In terms of the single-KP-root phonon factor $\mathcal{Z}(q;p) := (p_{\mathrm{ph},b}(q) - p)/(p_{\mathrm{ph},a}(q) - p)$, a defect at KP vertex $q_D = -s/(2\sqrt{3})$ ($s = \eta\sigma$) has $z_{\mathrm{ph}}(q; q_D, \eta) = \mathcal{Z}(q; q_D)^{\eta}$, while a soliton on edge (p_a, p_b) has $A(\mathrm{sol}(p_a, p_b), \mathrm{ph}(q)) = \mathcal{Z}(q; p_b)/\mathcal{Z}(q; p_a)$. Setting $q_D = -(\cos\gamma)/\sqrt{3}$ ($\gamma = \arccos(s/2)$), the defect phase-shift derivative decomposes into a *2-root* $(-\eta, +\eta)$ *dipole* $\theta_D'(q; s, \eta) = \mathrm{Re}\sum_{s'=1}^{2} c_{D,s'} \Theta'(q; \psi_{s'})$ over the universal elementary kernel $\Theta'(q;\psi)$ of `oneloop_phase.pdf`, where $\sin\psi_1 = q_+ - q_D$ and $\sin\psi_2 = q_D - q_-$ are the two root differences between q_D and the remaining two vertices $q_\pm$ of its KP triple. Subtracting the two defect dipoles at p_b and p_a yields $(+1, -1, 0) - (0, +1, -1) = (+1, -2, +1)$, reproducing the 3-root soliton kernel $\theta_S'(q; \alpha, \epsilon)$.
3. **Closed-form algebraic solution of the defect vacuum growing-mode spectrum.** The 4×4 linearised sewing determinant for localised modes $\omega = \mathrm{i}\gamma_{\mathrm{grow}}$ factors in $S = \kappa_1 + \kappa_2 = \sqrt{1 + 2\gamma_{\mathrm{grow}}}$ into two quadratics $P_\mp(S; \eta\sigma)$ with perfect-square discriminants $\big(\frac{\sqrt{3}\pm 1}{2}\big)^2 (4 - \sigma^2)$, giving the exact eigenvalues
$$\omega_{1,2}(\sigma, \eta) = \mathrm{i}\,\frac{\eta\sigma\sqrt{4-\sigma^2} \pm \sqrt{3}(2-\sigma^2)}{8},$$
which explain analytically the thresholds $s = -1$ (where ω_1 emerges from $\omega = 0$ with a -2π Levinson jump), $s = +1$ (where ω_2 emerges from $\omega = 0$ with a second -2π Levinson jump), and $s = +2$ (where ω_1, ω_2 collide at $\mathrm{i}\sqrt{3}/4$ and split into a complex-conjugate oscillatory pair).
4. **Closed-form 3×2 dilogarithm formula, (E, P) absorption balance, and KP-triangle closure.** Because the static defect has no translational zero mode ($\theta_D^{\mathrm{phys}}(0^\pm) = 0$), the raw and antisymmetrised Faddeev–Korepin integrals coincide ($I_{SD} = J_{SD}^{\mathrm{phys}}$) and evaluate in closed form as a 3×2 sum of Legendre χ_2 functions. Across every

soliton absorption process $D(q_i) + S_i \to D(q_{i+1})$ ($S_i = \{q_i, q_{i+1}\}$), the one-loop defect vacuum energy and momentum shifts satisfy exact balance $(\Delta E_1^{\rm ren}(S_i), \Delta P_1(S_i)) = \rho_i\big(\Delta E_D^{\rm ren}(q_{i+1}) - \Delta E_D^{\rm ren}(q_i),\ \Delta P_D(q_{i+1}) - \Delta P_D(q_i)\big)$, and the one-loop phases obey exact defect fusion $I_{SD}(3; q_{i+1}, \eta) - I_{SD}(3; q_i, \eta) = \rho_i\, I_{SS}(3, S_i)$ with the intrinsic absorption sign $\rho_i = \eta\,\mathrm{sgn}(q_i - q_{i+1}) \in \{+1, -1\}$ ($\rho_{\rm tree} = \epsilon_3 \rho_i$). Consequently, around every closed KP triangle $q_1 \to q_2 \to q_3 \to q_1$, the raw Faddeev–Korepin one-loop two-soliton phase I_{SS} is an exact coboundary ($\sum_{i=1}^{3} \rho_i\, I_{SS}(3, S_i) = 0$), whereas adding the $+\frac{\pi}{2}\delta_{\epsilon_3, \epsilon_{S_i}}$ zero-mode boundary constant of J_{SS} (or the $\pm\frac{\pi}{2}$ jump of the Γ-ansatz) leaves a non-zero coboundary defect $\frac{\pi}{2}\sum_{i=1}^{3} \rho_i\, \delta_{\epsilon_3, \epsilon_{S_i}} \in \{0, \pm\frac{\pi}{2}\}$.

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1 Setup, conventions, and the Route E question at one loop

We work in the fixed project conventions (`README.md`, `defect_semiclassics.pdf`, `oneloop_phase.pdf`) for the **good** Boussinesq equation **(a)**:

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \varphi_x = 2(\ln\tau)_{xx}, \tag{1}$$

with bulk Lagrangian density and canonical symplectic form

$$\mathcal{L} = \frac{1}{2}\varphi_t^2 - \frac{1}{2}\varphi_x^2 + \varphi_x^3 - \frac{1}{2}\varphi_{xx}^2, \qquad \Omega = \int_{-\infty}^{\infty} \mathrm{d}\pi \wedge \mathrm{d}\varphi, \quad \pi = \varphi_t, \tag{2}$$

in which $\hbar$ is normalised. The Corrigan–Zambon (CZ) potential $u_{\rm CZ}$ of [1] is related to φ by the explicit dictionary **(c)** `[defect_semiclassics.pdf]`:

$$u_{\rm CZ} = -3\varphi = -6(\ln\tau)_x, \qquad u = -\frac{1}{3}(u_{\rm CZ})_x, \qquad \mathcal{L}_{\rm CZ} = 9\,\mathcal{L}. \tag{3}$$

One-soliton and KP-root data. A one-soliton of (1) with amplitude parameter $\alpha \in (0, \pi/2)$ and direction sign $\epsilon \in \{+1, -1\}$ has **(a)**

$$k = \sin\alpha, \quad v = \epsilon\cos\alpha, \quad \omega = kv, \quad \ell = -\epsilon\cos 3\alpha, \quad M = 2k, \quad \Pi = -Mv, \quad P = \frac{2}{3}\epsilon\, k^3 \cos\alpha. \tag{4}$$

Its two KP roots **(a,c)**

$$p_a(\alpha,\epsilon) = \frac{k + v/\sqrt{3}}{2} = \frac{1}{\sqrt{3}}\cos\Big(\alpha - \frac{\epsilon\pi}{6}\Big), \qquad p_b(\alpha,\epsilon) = \frac{v/\sqrt{3} - k}{2} = -\frac{1}{\sqrt{3}}\cos\Big(\alpha + \frac{\epsilon\pi}{6}\Big) \tag{5}$$

are two of the three roots $\{q_1, q_2, q_3\}$ of the depressed KP cubic

$$p^3 - \frac{1}{4}p = \frac{\ell}{4\sqrt{3}} = -\frac{\epsilon}{4\sqrt{3}}\cos 3\alpha, \qquad q_1 + q_2 + q_3 = 0, \quad q_1q_2 + q_2q_3 + q_3q_1 = -\frac{1}{4}, \tag{6}$$

ordered as $p_a > p_b$ so that $k = p_a - p_b > 0$ and $v = \sqrt{3}(p_a + p_b)$. The three roots of (6) are

$$q_r(\alpha,\epsilon) = \frac{1}{\sqrt{3}}\cos\Big(\phi_r^{(\epsilon)} + \frac{\pi}{6}\Big) = -\frac{1}{\sqrt{3}}\sin\Big(\phi_r^{(\epsilon)} - \frac{\pi}{3}\Big), \tag{7}$$

and the three root differences are $\sin\phi_r^{(\epsilon)}$ $(r = 1, 2, 3)$, where

$$(\phi_1, \phi_2, \phi_3) := \Big(\alpha + \frac{2\pi}{3},\ \alpha - \frac{2\pi}{3},\ \alpha\Big), \qquad \phi_r^{(\epsilon)} := \begin{cases} \phi_r, & \epsilon = +1, \\ \pi - \phi_r, & \epsilon = -1, \end{cases} \qquad (c_1, c_2, c_3) := (1, 1, -2). \tag{8}$$

The Corrigan–Zambon type-I defect and its KP vertex. In CZ variables $u_{\mathrm{CZ}} = -3\varphi$ $(x < 0)$ and $v_{\mathrm{CZ}} = -3\psi$ $(x > 0)$, the type-I integrable defect at $x = 0$ is defined by four sewing conditions (CZ eqs. (2.4), (3.15)–(3.18) with $\alpha_{\mathrm{CZ}} = 1/(2\sqrt{3})$, $\beta_{\mathrm{CZ}} = \eta/(2\sqrt{3})$, $\eta \in \{+1, -1\}$, and real defect parameter $\sigma \in \mathbb{R}$) **(a)** [1]. Defining the signed defect parameter and KP vertex **(c)** `[defect_semiclassics.pdf]`

$$s := \eta\sigma \in \mathbb{R}, \qquad q_D := -\frac{s}{2\sqrt{3}} = -\frac{\eta\sigma}{2\sqrt{3}}, \tag{9}$$

a one-soliton $\tau_L = 1 + E_1$ $(x < 0)$, $\tau_R = 1 + z_1E_1$ $(x > 0)$ with $E_1 = \mathrm{e}^{k_1(x - v_1t - x_0)}$ crosses the defect without reflection and acquires the left-to-right Hirota transmission factor **(a,c)** [1]

$$z_1(\alpha_1, \epsilon_1; q_D, \eta) = \left(\frac{k_1 - v_1/\sqrt{3} - s/\sqrt{3}}{k_1 + v_1/\sqrt{3} + s/\sqrt{3}}\right)^{-\eta} = \left(\frac{p_{1,b} - q_D}{p_{1,a} - q_D}\right)^{\eta}. \tag{10}$$

For a soliton of direction $\epsilon_1 = \pm 1$ incident from $x \to -\epsilon_1\infty$ and transmitted to $x \to +\epsilon_1\infty$, the directional transmission factor is $z_t = z_1^{\epsilon_1} = \big((p_{1,b} - q_D)/(p_{1,a} - q_D)\big)^{\epsilon_1\eta}$ and the spatial shift is $\Delta x_D = -k_1^{-1}\ln|z_t|$.

What this write-up resolves (Route E, step 2). In Phase 2d (`defect_semiclassics.pdf`) it was shown that at tree level $(O(\hbar^{-1}))$, a defect sits at a single KP vertex q_D, a soliton $S = \{p_a, p_b\}$ is a chord between two vertices of a KP triple $\{q_1, q_2, q_3\}$, and absorbing $S_i = \{q_i, q_{i+1}\}$ shifts the defect vertex $q_i \to q_{i+1}$ while shifting the tree-level transmission phase of any test soliton 3 by the two-soliton phase $\delta_0(3, S_i) + M_{S_i}\Pi_3$ up to a sign. Meanwhile, in Phase 2e (`oneloop_phase.pdf`), we computed the exact one-loop $(O(\hbar^0))$ two-soliton phase directly from the continuum phonon phase shifts and found that the raw Faddeev–Korepin integral I_{12} and its antisymmetrised partner J_{12} differ only by a direction-dependent constant $+\frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}$ from the

translational zero-mode jump $\theta_S(0^+) - \theta_S(0^-) = -2\pi$, whereas both have zero head-on constant and rule out the Γ-ansatz for head-on collisions.

Here we carry out **Option 1 (Route E, step 2)**: we compute the exact one-loop soliton–defect transmission phase on the CZ defect, solve the defect vacuum stability spectrum in closed algebraic form, establish exact one-loop energy–momentum balance in soliton absorption, and test KP-triangle defect fusion at one loop. We show that one-loop defect fusion holds identically across every KP absorption edge and closes around every KP triangle *if and only if* the one-loop two-soliton phase uses the raw Faddeev–Korepin integral I_{12} (without the $+\frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}$ boundary constant of J_{12}).

2 Exact time-dependent phonon eigenmodes on N-soliton + defect backgrounds

2.1 Three-soliton Hirota solution on the CZ defect

Corrigan and Zambon gave explicit Hirota formulas for 1- and 2-soliton solutions on the type-I defect (CZ eqs. (5.3), (5.6)) **(a)** [1]. By the standard Hirota dressing of Bäcklund/defect transformations, the N-soliton solution on the defect is obtained by replacing each soliton weight $E_j = \mathrm{e}^{k_j x - \omega_j t + \delta_j}$ in $\tau_L(E_1, \dots, E_N)$ $(x < 0)$ by $z_j E_j$ in $\tau_R(z_1 E_1, \dots, z_N E_N)$ $(x > 0)$, with the bulk interaction coefficients $A_{ij} = A(i, j)$ unchanged. For $N = 3$,

$$\tau_L = 1 + \sum_{j=1}^{3} E_j + \sum_{1 \le i < j \le 3} A_{ij} E_i E_j + A_{12}A_{13}A_{23}\, E_1 E_2 E_3, \qquad \tau_R(E_1, E_2, E_3) = \tau_L(z_1 E_1, z_2 E_2, z_3 E_3). \tag{11}$$

Substituting $u_{\mathrm{CZ}} = -6(\ln \tau_L)_x$ and $v_{\mathrm{CZ}} = -6(\ln \tau_R)_x$ into the four full nonlinear CZ sewing conditions at $x = 0$ (CZ eqs. (3.15)–(3.18) in `verify_defect_solutions.py`), we verify at generic parameters that all four nonlinear sewing conditions hold identically at 35-digit precision (maximum residual 1.20×10^{-34}) **(c)** `[oneloop_defect_scattering_and_stability.py]`.

2.2 Time-dependent phonon scattering on 1- and 2-soliton + defect backgrounds

To obtain the exact continuum phonon eigenmodes on the defect vacuum ($N = 0$), the 1-soliton + defect background ($N = 1$), and the 2-soliton + defect background ($N = 2$), we analytically continue the last soliton parameter to a dispersive plane wave **(c)** `[oneloop_defect_scattering_and_stability.py`

$$k_{\mathrm{ph}} = \mathrm{i}q, \qquad \omega_{\mathrm{ph}} = \mathrm{i}\omega(q), \qquad \omega(q) := |q|\sqrt{1+q^2}, \qquad v_{\mathrm{ph}}(q) = \frac{\omega(q)}{q} = \mathrm{sgn}(q)\sqrt{1+q^2} \qquad (q \in \mathbb{R}\backslash\{0\}), \tag{12}$$

and expand $\varphi_{L,R}(x,t) = 2\partial_x(\ln \tau_{L,R}) = \varphi^{(0)}_{L,R}(x,t) + \varepsilon\, \delta\varphi_{L,R}(x,t;q) + O(\varepsilon^2)$ to first order in $E_{\mathrm{ph}} = \varepsilon\, \mathrm{e}^{\mathrm{i}(qx - \omega(q)t)}$.

Claim 2.1 (Exact reflectionless phonon eigenmodes on N-soliton + defect backgrounds, (c)). `[oneloop_defect_scattering_and_stability.py]` *For $N \in \{0, 1, 2\}$ solitons cross-sewing the CZ defect at $x = 0$:*

1. *The linearised Hirota field $(\delta\varphi_L(x,t;q), \delta\varphi_R(x,t;q))$ satisfies the bulk linearised good Boussinesq equation*

$$\big[\partial_t^2 - \partial_x^2 + \partial_x^4 + 6\partial_x(\varphi^{(0)}_{L,R})_x \partial_x\big]\delta\varphi_{L,R}(x,t;q) = 0$$

for $x < 0$ and $x > 0$, together with all four linearised CZ sewing conditions at $x = 0$ for every $t \in \mathbb{R}$ (verified at 35-digit precision with maximum sewing residual 3.94×10^{-34} for $N = 1$ and 2.10×10^{-34} for $N = 2$ across interacting $t = 0.15$ and asymptotic $t = \pm 15$ times).

2. *Away from the soliton cores and the defect ($|x| \to \infty$), the eigenmode is purely oscillatory with* ***zero reflection*** *($R(q) = 0$) and* ***zero asymptotic evanescent waves:***

$$\delta\varphi_L(x,t;q) \underset{x\to-\infty}{\sim} 2\mathrm{i}q\,\mathrm{e}^{\mathrm{i}(qx-\omega(q)t)}, \qquad \delta\varphi_R(x,t;q) \underset{x\to+\infty}{\sim} 2\mathrm{i}q\,T_{\mathrm{tot}}(q)\,\mathrm{e}^{\mathrm{i}(qx-\omega(q)t)}, \tag{13}$$

where the total left-to-right transmission amplitude factors identically for all $t \in \mathbb{R}$ into the product of the individual soliton and defect transmission amplitudes:

$$T_{\mathrm{tot}}(q) = \left(\prod_{j=1}^{N} A\big(j,\mathrm{ph}(q)\big)\right) z_{\mathrm{ph}}(q;q_D,\eta), \qquad |T_{\mathrm{tot}}(q)| = 1. \tag{14}$$

Here $A(j,\mathrm{ph}(q)) = \mathrm{e}^{\mathrm{i}\theta_S(q;\alpha_j,\epsilon_j)}$ is the bulk soliton–phonon transmission amplitude of `oneloop_phase.pdf` *and*

$$z_{\mathrm{ph}}(q;q_D,\eta) = \left(\frac{\mathrm{sgn}(q)\frac{\sqrt{1+q^2}}{\sqrt{3}} + \frac{s}{\sqrt{3}} - \mathrm{i}q}{\mathrm{sgn}(q)\frac{\sqrt{1+q^2}}{\sqrt{3}} + \frac{s}{\sqrt{3}} + \mathrm{i}q}\right)^{\eta} = \left(\frac{\frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} - q_D - \frac{\mathrm{i}q}{2}}{\frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} - q_D + \frac{\mathrm{i}q}{2}}\right)^{\eta} =: \mathrm{e}^{\mathrm{i}\theta_D(q;s,\eta)} \tag{15}$$

is the defect–phonon transmission amplitude **(c)** `[defect_transmission_phase.py, oneloop_defect_sc`

Consequence for the vacuum-subtracted one-loop phase. On the 1-soliton + defect background, the total phonon phase shift is the sum $\theta_{\mathrm{tot}}(q) = \theta_S(q;\alpha,\epsilon) + \theta_D(q;s,\eta)$, independent of time t. Subtracting the defect vacuum ($N = 0$, phase shift $\theta_D(q;s,\eta)$) and the isolated soliton ($N = 1$ without defect, phase shift $\theta_S(q;\alpha,\epsilon)$) leaves only the bilinear cross-interaction between $\theta_S(q;\alpha,\epsilon)$ and $\theta_D(q;s,\eta)$ in the Faddeev–Korepin trace formula (Section 5). Likewise, on the 2-soliton + defect background, $\theta_{\mathrm{tot}}(q) = \theta_S(q;\alpha_1,\epsilon_1) + \theta_S(q;\alpha_2,\epsilon_2) + \theta_D(q;s,\eta)$, so the one-loop 2-soliton + defect phase is the exact sum of pairwise phases $\delta^{(1)}_{\mathrm{tot}}(1,2;D) = \delta^{(1)}_{12} + \delta^{(1)}_{1D} + \delta^{(1)}_{2D}$ with zero three-body irreducible one-loop correction **(c)**.

3 KP-vertex factorisation and the 2-root dipole decomposition of $\theta'_D(q;s,\eta)$

3.1 Single-KP-root phonon factor $\mathcal{Z}(q;p)$

Under the phonon continuation (12), the two KP roots (5) become the complex-conjugate pair **(c)**

$$p_{\mathrm{ph},a}(q) := \frac{\mathrm{sgn}(q)\frac{\sqrt{1+q^2}}{\sqrt{3}} + \mathrm{i}q}{2}, \qquad p_{\mathrm{ph},b}(q) := \frac{\mathrm{sgn}(q)\frac{\sqrt{1+q^2}}{\sqrt{3}} - \mathrm{i}q}{2} = \overline{p_{\mathrm{ph},a}(q)}. \tag{16}$$

For any real or complex KP root p, define the elementary vertex–phonon ratio **(c)** `[oneloop_defect_scattering_`

$$\mathcal{Z}(q;p) := \frac{p_{\mathrm{ph},b}(q) - p}{p_{\mathrm{ph},a}(q) - p} = \frac{\frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} - p - \frac{\mathrm{i}q}{2}}{\frac{\mathrm{sgn}(q)\sqrt{1+q^2}}{2\sqrt{3}} - p + \frac{\mathrm{i}q}{2}}. \tag{17}$$

Claim 3.1 (KP-vertex/chord factorisation of phonon scattering, (c)). `[oneloop_defect_scattering_`
For every $q \in \mathbb{R}\setminus\{0\}$, every defect vertex q_D, and every soliton $S = \{p_a, p_b\}$, the defect and soliton phonon transmission amplitudes factor through $\mathcal{Z}(q;p)$ as

$$z_{\mathrm{ph}}(q;q_D,\eta) = \mathcal{Z}(q;q_D)^{\eta}, \qquad A\big(\mathrm{sol}(p_a,p_b),\mathrm{ph}(q)\big) = \frac{\mathcal{Z}(q;p_b)}{\mathcal{Z}(q;p_a)} = z_{\mathrm{ph}}(q;p_b,+1)\,z_{\mathrm{ph}}(q;p_a,-1). \tag{18}$$

Consequently, the soliton–phonon phase shift is the exact difference of two defect–phonon phase shifts at the endpoints p_b and p_a of the soliton chord (verified to 6.48×10^{-36}):

$$\theta'_S(q;\alpha,\epsilon) = \theta'_D\big(q; s(p_b), +1\big) - \theta'_D\big(q; s(p_a), +1\big), \qquad s(p) := -2\sqrt{3}\,p. \tag{19}$$

3.2 The 2-root dipole decomposition of $\theta'_D(q;s,\eta)$

In Phase 2e (`oneloop_phase.pdf`, Claim 2.2) we showed that the soliton–phonon phase-shift derivative $\theta'_S(q;\alpha,\epsilon)$ decomposes into a sum over the three roots of its KP cubic with weights $(c_1,c_2,c_3) = (1,1,-2)$:

$$\theta'_S(q;\alpha,\epsilon) = \sum_{r=1}^{3} c_r\,\Theta'\Big(q;\phi_r^{(\epsilon)}\Big), \qquad \Theta'(q;\psi) := \frac{\sin\psi}{q^2+\sin^2\psi} + \frac{\operatorname{sgn}(q)}{\sqrt{1+q^2}}\frac{\sin\psi\cos\psi}{q^2+\sin^2\psi}. \tag{20}$$

Why did a soliton have weights $(1,1,-2)$? Claim 3.2 below reveals the reason: each single KP vertex q_D is a *2-root dipole* with weights $(-1,+1)$ associated with the two chords connecting q_D to the other two vertices $q_\pm$ of its KP cubic, and subtracting the two dipoles at p_b and p_a gives $(+1,-1,0) - (0,+1,-1) = (+1,-2,+1)$!

Claim 3.2 (2-root dipole decomposition of the defect phase-shift derivative, (c)). `[oneloop_defect_scattering_and_stability.py]` *Given $s = \eta\sigma \in \mathbb{R}$ and $q_D = -s/(2\sqrt{3})$, define*

$$\gamma := \arccos\Big(\frac{s}{2}\Big), \qquad \psi_1 := \frac{2\pi}{3} - \gamma, \qquad \psi_2 := -\frac{2\pi}{3} - \gamma, \qquad (c_{D,1}, c_{D,2}) := (-\eta, +\eta), \tag{21}$$

where for $|s| \le 2$ ($|q_D| \le 1/\sqrt{3}$) the angle $\gamma \in [0,\pi]$ is real, while for $|s| > 2$ (γ complex) $\psi_{1,2}$ are complex conjugates modulo π. Then:

1. *The vertex $q_D = -(\cos\gamma)/\sqrt{3}$ belongs to the KP cubic $4q^3 - q = 4q_D^3 - q_D = -\frac{\cos 3\gamma}{3\sqrt{3}}$, whose other two roots are $q_\pm = \frac{-q_D \pm \sqrt{1-3q_D^2}}{2} = \frac{-q_D \pm \sin\gamma}{2}$, and the two root differences from q_D are identically*

$$\sin\psi_1 = \sin\Big(\gamma + \frac{\pi}{3}\Big) = q_+ - q_D, \qquad \sin\psi_2 = \sin\Big(\gamma - \frac{\pi}{3}\Big) = q_D - q_-. \tag{22}$$

2. *For every $q \in \mathbb{R}\setminus\{0\}$ and every $s \in \mathbb{R}$ (both $|s| \le 2$ and $|s| > 2$), the defect phase-shift derivative $\theta'_D(q;s,\eta) := -\mathrm{i}\partial_q \ln z_{\mathrm{ph}}(q;q_D,\eta)$ is given by the 2-root dipole sum (verified to 1.88×10^{-36}):*

$$\theta'_D(q;s,\eta) = -\eta\,\frac{2\,\operatorname{sgn}(q)\frac{q}{2\sqrt{3}\sqrt{1+q^2}}\frac{q}{2} + \Big(\operatorname{sgn}(q)\frac{\sqrt{1+q^2}}{2\sqrt{3}} - q_D\Big)}{\Big(\operatorname{sgn}(q)\frac{\sqrt{1+q^2}}{2\sqrt{3}} - q_D\Big)^2 + \frac{q^2}{4}} = \operatorname{Re}\sum_{s'=1}^{2} c_{D,s'}\,\Theta'(q;\psi_{s'}). \tag{23}$$

Derivation. Rationalising the denominator of (23),

$$\Big(\operatorname{sgn}(q)\frac{\sqrt{1+q^2}}{2\sqrt{3}} - q_D\Big)^2 + \frac{q^2}{4} = \frac{1+4q^2+3s^2+2s\,\operatorname{sgn}(q)\sqrt{1+q^2}}{12}.$$

Multiplying numerator and denominator by $(1+4q^2+3s^2) - 2s\,\operatorname{sgn}(q)\sqrt{1+q^2}$ produces

$$(1+4q^2+3s^2)^2 - 4s^2(1+q^2) = 16\Big(q^2+\sin^2\Big(\gamma+\frac{\pi}{3}\Big)\Big)\Big(q^2+\sin^2\Big(\gamma-\frac{\pi}{3}\Big)\Big) = 16(q^2+\sin^2\psi_1)(q^2+\sin^2\psi_2).$$

Partial-fraction decomposition in q^2 then splits the rational part into $\frac{-\sin\psi_1}{q^2+\sin^2\psi_1} + \frac{\sin\psi_2}{q^2+\sin^2\psi_2}$ and the $\operatorname{sgn}(q)/\sqrt{1+q^2}$ part into $\frac{-\sin\psi_1\cos\psi_1}{q^2+\sin^2\psi_1} + \frac{\sin\psi_2\cos\psi_2}{q^2+\sin^2\psi_2}$, which is precisely (23). □

4 Closed-form algebraic solution of the defect vacuum growing-mode spectrum

In `corrigan-zambon/defect_vacuum_stability.py`, the linearised stability of the defect vacuum $\varphi_L^{(0)} = \varphi_R^{(0)} = 0$ was studied numerically by searching for roots of a 4×4 sewing determinant, revealing a threshold at $s = \eta\sigma = -1$, a second threshold at $s = +1$, and a mode collision at $s = +2$. Using the KP-vertex structure, we now factor the 4×4 determinant in closed algebraic form **(c)** `[oneloop_defect_scattering_and_stability.py]`.

Claim 4.1 (Closed-form factorisation of the localised-mode determinant and exact growing-mode eigenvalues, (c)). `[oneloop_defect_scattering_and_stability.py]` *Consider a localised eigenmode of frequency $\omega = \mathrm{i}\gamma_{\mathrm{grow}}$ ($\gamma_{\mathrm{grow}} > 0$), which decays as $\mathrm{e}^{+\kappa_j x}$ for $x < 0$ and $\mathrm{e}^{-\kappa_j x}$ for $x > 0$, where $\kappa_{1,2}^2 = \frac{1 \pm \sqrt{1-4\gamma_{\mathrm{grow}}^2}}{2}$ satisfy $\kappa_1\kappa_2 = \gamma_{\mathrm{grow}}$ and*

$$S := \kappa_1 + \kappa_2 = \sqrt{1 + 2\gamma_{\mathrm{grow}}} > 1 \qquad \Longleftrightarrow \qquad \gamma_{\mathrm{grow}} = \frac{S^2 - 1}{2}. \tag{24}$$

Then:

1. *After dividing out the trivial root $\kappa_1 = \kappa_2$, the 4×4 linearised CZ sewing determinant factors in closed form as (verified to 3.89×10^{-36}):*

$$\det M_{\mathrm{red}}(S;\sigma,\eta) := \frac{\det M_{4\times4}(\gamma_{\mathrm{grow}};\sigma,\eta)}{(\kappa_1-\kappa_2)^2} = -\frac{S^2(S^2-1)}{24}\, P_-(S;\eta\sigma)\, P_+(S;\eta\sigma), \tag{25}$$

   where $P_\mp(S;s)$ ($s = \eta\sigma$) are the two quadratic polynomials

$$P_\mp(S;s) := S^2 - \frac{\sqrt{3} \mp 1}{2}\, s\, S + \frac{s^2 - 2 \mp \sqrt{3}}{2}. \tag{26}$$

2. *The discriminants of $P_\mp(S;s)$ are perfect squares times $(4 - s^2)$:*

$$\Delta_\mp(s) = \left(\frac{\sqrt{3} \pm 1}{2}\right)^2 (4 - s^2), \tag{27}$$

   yielding the four roots

$$S_{-,\pm}(s) = \frac{(\sqrt{3}-1)s \pm (\sqrt{3}+1)\sqrt{4-s^2}}{4}, \qquad S_{+,\pm}(s) = \frac{(\sqrt{3}+1)s \pm (\sqrt{3}-1)\sqrt{4-s^2}}{4}. \tag{28}$$

3. *Substituting $S_{-,-}(s)$ and $S_{+,+}(s)$ into $\gamma_{\mathrm{grow}} = (S^2-1)/2$ gives the **exact closed-form localised defect eigenvalues** (verified against numerical root-finding of $\det M_{4\times4} = 0$ to 1.62×10^{-35}):*

$$\boxed{\omega_1(s) = \mathrm{i}\,\frac{s\sqrt{4-s^2} - \sqrt{3}(2-s^2)}{8}, \qquad \omega_2(s) = \mathrm{i}\,\frac{s\sqrt{4-s^2} + \sqrt{3}(2-s^2)}{8} \qquad (s = \eta\sigma).} \tag{29}$$

Complete analytic stability phase diagram and Levinson mode count (c). `[oneloop_defect_scatter`
Formula (25)–(29) accounts analytically for every regime of the CZ defect vacuum:

- **Strictly stable half-line $s \le -1$ ($q_D \ge \frac{1}{2\sqrt{3}}$):** Neither $S_{-,-}(s)$ nor $S_{+,+}(s)$ exceeds 1, so there are **zero** localised growing or bound modes. For $s \in [-2,-1)$, $\gamma \in (2\pi/3, \pi]$, so $\psi_1 = 2\pi/3 - \gamma \in [-\pi/3, 0)$ has $\sin\psi_1 < 0$ while $\psi_2 = -2\pi/3 - \gamma \in [-5\pi/3, -4\pi/3)$ has $\sin\psi_2 > 0$. Since $\int_{-\infty}^{\infty} \Theta'(q;\psi)\,\mathrm{d}q = 2\pi\,\mathrm{sgn}(\sin\psi)$ is an even function of q while the $\mathrm{sgn}(q)/\sqrt{1+q^2}$ part is odd and integrates to $\pm\frac{2\pi}{3}$ on $q>0$ and $q<0$ separately, the total Levinson phase winding across $q \in (-\infty, \infty)$ is

$$\Delta\theta_D(s, \eta = 1) := \int_{-\infty}^{\infty} \theta'_D(q; s, 1)\,\mathrm{d}q = \pi\big(-\mathrm{sgn}(\sin\psi_1) + \mathrm{sgn}(\sin\psi_2)\big) = \begin{cases} +2\pi, & s < -1, \\ 0, & -1 < s < +1, \\ -2\pi, & s > +1, \end{cases} \tag{30}$$

  decreasing by 2π each time a growing mode bifurcates out of $\omega = 0$.

- **First threshold $s = -1$ ($q_D = +\frac{1}{2\sqrt{3}}$, $\gamma = 2\pi/3$):** Here $\psi_1 = 0$, so $\sin\psi_1$ changes sign from negative to positive, $P_-(1;-1) = 0$ ($S_{-,-}(-1) = 1$, $\omega_1(-1) = 0$), and the first growing mode $\omega_1(s) = \mathrm{i}\gamma_1(s)$ bifurcates out of the $\omega = 0$ continuum edge for $s > -1$, accompanied by a jump of -2π in $\Delta\theta_D$ (from $+2\pi$ to 0).

- **Second threshold $s = +1$ ($q_D = -\frac{1}{2\sqrt{3}}$, $\gamma = \pi/3$):** Here $\psi_2 = -\pi$, so $\sin\psi_2$ changes sign from positive to negative, $P_+(1;+1) = 0$ ($S_{+,+}(+1) = 1$, $\omega_2(+1) = 0$), and the second growing mode $\omega_2(s) = \mathrm{i}\gamma_2(s)$ bifurcates out of $\omega = 0$ for $s > +1$, accompanied by a second jump of -2π in $\Delta\theta_D$ (from 0 to -2π).

- **Collision threshold $s = +2$ ($q_D = -1/\sqrt{3}$, $\gamma = 0$) and oscillatory regime $s > 2$:** At $s = +2$, the discriminant $4 - s^2$ vanishes and the two pure-imaginary eigenvalues collide at $\omega_1(2) = \omega_2(2) = \mathrm{i}\sqrt{3}/4$ ($S = (\sqrt{3}+1)/2$). For $s > 2$, $\sqrt{4-s^2} = \pm\mathrm{i}\sqrt{s^2-4}$ and (29) becomes the complex-conjugate oscillatory unstable pair

$$\omega_\pm(s) = \frac{\pm s\sqrt{s^2-4} + \mathrm{i}\sqrt{3}(s^2-2)}{8}.$$

Why localised defect modes cancel in the soliton–defect transmission phase (c). In the strictly stable regime $s \le -1$, no localised defect modes exist at all. Even when $s > -1$, a soliton crosses the defect at $x = 0$ in a finite time interval and leaves the asymptotic defect state at $t \to +\infty$ identical to that at $t \to -\infty$ (unless $q_D = p_{a,b}$, which is an inelastic absorption threshold). Because the one-loop soliton–defect transmission phase is the vacuum-subtracted difference between the soliton+defect sector and the isolated-soliton plus defect-vacuum sectors, any localised mode attached to the static defect has the same eigenfrequency before and after the collision and cancels between the soliton+defect and defect-vacuum functional determinants.

5 Closed-form 3×2 dilogarithm formula for the one-loop soliton–defect phase and (E, P) absorption balance

5.1 Equality of raw and antisymmetrised Faddeev–Korepin integrals: $I_{SD} = J_{SD}^{\mathrm{phys}}$

In Phase 2e (`oneloop_phase.pdf`), the raw Faddeev–Korepin integral I_{12} and the antisymmetrised integral J_{12} for two solitons differed by $J_{12} - I_{12} = \frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}$ because the soliton has two discrete zero modes (translational and α-derivative), giving $\theta_S(0^+) - \theta_S(0^-) = -2\pi$ and a non-vanishing integration-by-parts boundary term at $q = 0^{\pm}$.

For a soliton scattering on a *static* defect at $x = 0$, the defect breaks translation invariance explicitly and has **no translational zero mode**. Indeed, from (15), as $q \to 0^{\pm}$,

$$z_{\rm ph}(0^{\pm}; q_D, \eta) = \left(\frac{\frac{\pm 1}{2\sqrt{3}} - q_D - \mathrm{i}\,0^{+}}{\frac{\pm 1}{2\sqrt{3}} - q_D + \mathrm{i}\,0^{+}} \right)^{\eta} = 1 \qquad \left(q_D \neq \pm \frac{1}{2\sqrt{3}} \right), \tag{31}$$

so the physical defect phase shift $\theta_D^{\rm phys}(q; s, \eta) := -\mathrm{i} \ln z_{\rm ph}(q; q_D, \eta)$ defined continuously from $q = 0^{\pm}$ vanishes at both long-wavelength limits:

$$\theta_D^{\rm phys}(0^{+}; s, \eta) = \theta_D^{\rm phys}(0^{-}; s, \eta) = 0, \qquad \theta_S(\pm\infty; \alpha, \epsilon) = 0. \tag{32}$$

Integrating by parts on $(-\infty, 0)$ and $(0, \infty)$ therefore yields zero boundary terms at both $q = 0^{\pm}$ and $q = \pm\infty$!

Claim 5.1 (Equality $I_{SD} = J_{SD}^{\rm phys}$ and closed-form 3×2 dilogarithm formula, (c)). `[oneloop_defect_phase_and_fusion.py]` *For every soliton $(\alpha, \epsilon) \in (0, \pi/2) \times \{+1, -1\}$ and every defect $(s, \eta) \in (\mathbb{R} \setminus \{\pm 1\}) \times \{+1, -1\}$, the raw and antisymmetrised Faddeev–Korepin one-loop soliton–defect continuum integrals coincide:*

$$\begin{aligned} I_{SD}(\alpha, \epsilon; s, \eta) &:= \frac{1}{4\pi} \int_{-\infty}^{\infty} \theta_S(q; \alpha, \epsilon)\, \theta_D'(q; s, \eta)\, \mathrm{d}q \\ &= \frac{1}{8\pi} \int_{-\infty}^{\infty} \Big[\theta_S(q; \alpha, \epsilon)\, \theta_D'(q; s, \eta) - \theta_D^{\rm phys}(q; s, \eta)\, \theta_S'(q; \alpha, \epsilon) \Big] \mathrm{d}q =: J_{SD}^{\rm phys}(\alpha, \epsilon; s, \eta), \end{aligned} \tag{33}$$

and evaluate in closed analytical form as the 3×2 dilogarithm sum (verified against numerical quadrature across $|s| < 2$ and $|s| > 2$ to 4.51×10^{-36}, and against the Faddeev–Korepin α-gradient $\partial_\alpha I_{SD} = \frac{1}{4\pi} \int (\partial_\alpha \theta_S) \theta_D'\, \mathrm{d}q$ to 9.40×10^{-38}):

$$\boxed{ I_{SD}(\alpha, \epsilon; s, \eta) = J_{SD}^{\rm phys}(\alpha, \epsilon; s, \eta) = \frac{\epsilon}{2\pi} \sum_{r=1}^{3} \sum_{s'=1}^{2} c_r\, c_{D,s'}\, \mathrm{Re}\, L\left(\tan\frac{\phi_r}{2} \left(\tan\frac{\psi_{s'}}{2} \right)^{-\epsilon} \right), } \tag{34}$$

where $(\phi_1, \phi_2, \phi_3) = (\alpha + 2\pi/3, \alpha - 2\pi/3, \alpha)$, $(c_1, c_2, c_3) = (1, 1, -2)$, $(\psi_1, \psi_2) = (2\pi/3 - \gamma, -2\pi/3 - \gamma)$ with $\gamma = \arccos(s/2)$, $(c_{D,1}, c_{D,2}) = (-\eta, +\eta)$, and

$$L(z) = \int_0^z \frac{2 \ln t}{1 - t^2}\, \mathrm{d}t = -\left[2\chi_2(z) + \ln z \, \ln\frac{1 - z}{1 + z} \right] \quad (\mathrm{Re}\, z > 0), \qquad L(-z) = -L(z), \tag{35}$$

with $\chi_2(z) = \frac{1}{2}\big(\mathrm{Li}_2(z) - \mathrm{Li}_2(-z) \big)$ the Legendre chi function. Moreover, I_{SD} obeys the exact parity/crossing and defect-orientation identities (verified to 2.26×10^{-36}):

$$I_{SD}(\alpha, -1; s, \eta) = I_{SD}(\alpha, +1; -s, \eta), \qquad I_{SD}(\alpha, \epsilon; s, -\eta) = -I_{SD}(\alpha, \epsilon; s, \eta). \tag{36}$$

5.2 One-loop defect vacuum shifts $(\Delta E_D^{\rm ren}, \Delta P_D)$ and exact absorption balance

Because the defect phase-shift derivative $\theta_D'(q; s, \eta)$ is a 2-root dipole (23) over the universal kernel $\Theta'(q; \psi)$ with $\sum_{s'=1}^{2} c_{D,s'} \sin\psi_{s'} \neq 0$, the raw zero-point energy shift of the defect vacuum has a UV divergence proportional to $\sum_{s'} c_{D,s'} \sin\psi_{s'}$ that is cancelled by the exact same bulk cubic tadpole counterterm as the soliton vacuum energy in `oneloop_phase.pdf` (since the defect sewing potential is linear in $u_{\rm CZ}$ and has zero bulk tadpole of its own). Meanwhile, the defect vacuum zero-point momentum shift ΔP_D is UV convergent without any counterterm because the odd part of $\theta_D'(q)$ decays as $O(q^{-3})$.

Claim 5.2 (Closed-form defect vacuum shifts and exact one-loop (E,P) balance in soliton absorption, (c)). `[oneloop_defect_phase_and_fusion.py]` *For any defect vertex $q_D = -s/(2\sqrt{3})$ and orientation $\eta \in \{+1,-1\}$:*

1. *The one-loop zero-point momentum $\Delta P_D(q_D,\eta) = \int_{-\infty}^{\infty} \frac{\mathrm{d}q}{4\pi}\, q\, \theta'_D(q;s,\eta)$ and normal-ordered zero-point energy $\Delta E_D^{\mathrm{ren}}(q_D,\eta)$ are given in closed form for all $s \in \mathbb{R}$ by (verified against numerical quadrature to 7.52×10^{-37}):*

$$\Delta P_D(q_D,\eta) = -\frac{1}{2\pi}\sum_{s'=1}^{2} c_{D,s'}\ \mathrm{Re}\left[\frac{1}{2}\sin\psi_{s'}\ln\frac{1-\cos\psi_{s'}}{1+\cos\psi_{s'}}\right], \tag{37}$$

$$\Delta E_D^{\mathrm{ren}}(q_D,\eta) = \frac{1}{2\pi}\sum_{s'=1}^{2} c_{D,s'}\ \mathrm{Re}\left[\sin\psi_{s'}\left(1-\frac{1}{2}\cos\psi_{s'}\ln\frac{1-\cos\psi_{s'}}{1+\cos\psi_{s'}}\right)\right]. \tag{38}$$

2. *Let $\{q_1,q_2,q_3\}$ be any KP triple ($|q_i| < 1/\sqrt{3}$), and let $S_i = \{q_i, q_{i+1}\}$ be the soliton connecting vertex q_i to vertex q_{i+1} ($q_4 \equiv q_1$). Define the intrinsic absorption sign*

$$\rho_i := \eta\ \mathrm{sgn}(q_i - q_{i+1}) \in \{+1,-1\}. \tag{39}$$

Then across every soliton absorption/emission step $D(q_i)+S_i \leftrightarrow D(q_{i+1})$, the one-loop soliton energy and momentum shifts $(\Delta E_1^{\mathrm{ren}}(S_i), \Delta P_1(S_i))$ equal the jump in the defect vacuum shifts (verified across 18 absorption edges to 3.20×10^{-36}):

$$\boxed{\Delta E_D^{\mathrm{ren}}(q_{i+1},\eta) - \Delta E_D^{\mathrm{ren}}(q_i,\eta) = \rho_i\, \Delta E_1^{\mathrm{ren}}(S_i), \qquad \Delta P_D(q_{i+1},\eta) - \Delta P_D(q_i,\eta) = \rho_i\, \Delta P_1(S_i).} \tag{40}$$

6 One-loop defect fusion, KP-triangle closure, and implications for the quantum S-matrix

6.1 Total one-loop soliton–defect phase at fixed physical momentum

Including the shift $\alpha \mapsto \alpha + \hbar\,\delta\alpha_1$ required to hold the physical soliton momentum $P_{\mathrm{phys}} = P(\alpha,\epsilon) + \hbar\,\Delta P_1(\alpha,\epsilon)$ fixed at $O(\hbar)$ (as in `oneloop_phase.pdf`, Section 3.3), where

$$\frac{\partial\delta_D^{\mathrm{trans}}(\alpha,\epsilon;s,\eta)}{\partial\alpha} = \Delta x_D(\alpha,\epsilon;s,\eta)\,\frac{\mathrm{d}P(\alpha,\epsilon)}{\mathrm{d}\alpha}, \qquad \Delta x_D(\alpha,\epsilon;s,\eta) = -\frac{\epsilon\eta}{k}\ln\left|\frac{p_b - q_D}{p_a - q_D}\right|, \tag{41}$$

gives the fixed-physical-momentum renormalisation shift

$$\delta_{D,\mathrm{renorm}}^{(q_D)}(\alpha,\epsilon;\eta) := -\frac{\partial\delta_D^{\mathrm{trans}}}{\partial P}\,\Delta P_1(\alpha,\epsilon) = -\Delta x_D(\alpha,\epsilon;s,\eta)\,\Delta P_1(\alpha,\epsilon), \tag{42}$$

where we follow the sign convention of `oneloop_defect_phase_and_fusion.py` in tracking $\Delta x_D\,\Delta P_1$.

6.2 Exact one-loop defect fusion and KP-triangle closure

We now test whether the tree-level defect fusion relation [1] (`defect_semiclassics.pdf`, Claim 5) survives at one loop.

Claim 6.1 (Exact one-loop defect fusion and KP-triangle coboundary closure, (c)). `[oneloop_defect_phase_and_fusion.py]` *Let $\{q_1,q_2,q_3\}$ be any KP triple, $S_i = \{q_i, q_{i+1}\}$ ($i = 1,2,3$) the three solitons on the edges of the KP triangle, $\rho_i = \eta\,\mathrm{sgn}(q_i - q_{i+1}) \in \{+1,-1\}$ the intrinsic absorption sign (39), and $3 = (\alpha_3,\epsilon_3)$ an arbitrary test soliton. Then:*

1. *At tree level, the directional transmission phase* $\delta_D^{\mathrm{trans}}(3;q_D,\eta)$ *obeys*

$$\delta_D^{\mathrm{trans}}(3;q_{i+1},\eta)-\delta_D^{\mathrm{trans}}(3;q_i,\eta)=\rho_{\mathrm{tree},i}\Big(\delta_0(3,S_i)+M_{S_i}\Pi_3\Big),\qquad \rho_{\mathrm{tree},i}=\epsilon_3\rho_i\in\{+1,-1\},\tag{43}$$

where the factor ϵ_3 *reflects the directional transmission exponent* $z_t=z_3^{\epsilon_3}$ *for a right-moving* ($\epsilon_3=+1$, $L\to R$) *versus left-moving* ($\epsilon_3=-1$, $R\to L$) *test soliton.*

2. *At one loop, both the Faddeev–Korepin continuum phase* $I_{SD}(3;q_D,\eta)$ *and the momentum-renormalisation shift* $\Delta x_D(3;q_D,\eta)\,\Delta P_1(\alpha_3,\epsilon_3)$ *satisfy* ***exact one-loop defect fusion*** *on every edge* $q_i\to q_{i+1}$ *(verified across* 12 *KP triangles* $=$ 36 *edges to* 4.51×10^{-36} *for* I_{SD} *and* 1.67×10^{-33} *for the momentum-renormalisation term):*

$$\boxed{I_{SD}(3;q_{i+1},\eta)-I_{SD}(3;q_i,\eta)=\rho_i\,I_{SS}(3,S_i),\qquad \Delta x_D(3;q_{i+1},\eta)-\Delta x_D(3;q_i,\eta)=\epsilon_3\rho_i\,\Delta X_3(3,S_i),}\tag{44}$$

where $I_{SS}(3,S_i)=I_{12}(\alpha_3,\epsilon_3;\alpha_{S_i},\epsilon_{S_i})$ *is the raw Faddeev–Korepin one-loop two-soliton phase of* `oneloop_phase.pdf` *and* $\Delta X_3(3,S_i)=-k_3^{-1}\ln|A_{12}(3,S_i)|$. *Equivalently, for the directional transmission integral* $I_{SD}^{\mathrm{trans}}(3;q_D,\eta):=\epsilon_3\,I_{SD}(3;q_D,\eta)$, *both terms in* (44) *carry the exact same sign* $\rho_{\mathrm{tree},i}=\epsilon_3\rho_i$ *as tree level* (43).

3. *Summing* (44) *around any closed KP triangle* $q_1\to q_2\to q_3\to q_1$, *the raw Faddeev–Korepin one-loop two-soliton phase* I_{SS} *and the momentum-renormalisation shift close* ***identically to zero*** *(verified across all* 12 *triangles to* 1.50×10^{-35} *and* 7.64×10^{-34}*):*

$$\sum_{i=1}^{3}\rho_i\,I_{SS}(3,S_i)=0,\qquad \sum_{i=1}^{3}\rho_{\mathrm{tree},i}\,\Delta X_3(3,S_i)\,\Delta P_1(\alpha_3,\epsilon_3)=0.\tag{45}$$

4. *By contrast, if one replaces* $I_{SS}(3,S_i)$ *by the antisymmetrised integral* $J_{SS}(3,S_i)=I_{SS}(3,S_i)+\frac{\pi}{2}\delta_{\epsilon_3,\epsilon_{S_i}}$ *(or by any candidate with a direction-dependent* $\pm\frac{\pi}{2}$ *constant, such as the* Γ*-ansatz), the closed KP triangle acquires a* ***non-zero coboundary obstruction*** *(verified to* 1.35×10^{-35}*):*

$$\sum_{i=1}^{3}\rho_i\,J_{SS}(3,S_i)=\frac{\pi}{2}\sum_{i=1}^{3}\rho_i\,\delta_{\epsilon_3,\epsilon_{S_i}}\in\Big\{0,\ +\frac{\pi}{2},\ -\frac{\pi}{2}\Big\}.\tag{46}$$

Algebraic proof of Claim 6.1(2)–(4). By Claim 3.1, the soliton–phonon phase-shift derivative for $S_i=\{p_{S_i,a},p_{S_i,b}\}$ ($p_{S_i,a}=\max(q_i,q_{i+1})$, $p_{S_i,b}=\min(q_i,q_{i+1})$) is the exact coboundary $\theta_S'(q;S_i)=\theta_D'(q;s(p_{S_i,b}),+1)-\theta_D'(q;s(p_{S_i,a}),+1)$. Since $\theta_D'(q;s,\eta)=\eta\,\theta_D'(q;s,+1)$ and $(q_{i+1},q_i)=(p_{S_i,b},p_{S_i,a})$ when $q_{i+1}<q_i$ ($\rho_i=+\eta$) and $(p_{S_i,a},p_{S_i,b})$ when $q_{i+1}>q_i$ ($\rho_i=-\eta$), we have pointwise for every $q\in\mathbb{R}\setminus\{0\}$:

$$\theta_D'(q;s(q_{i+1}),\eta)-\theta_D'(q;s(q_i),\eta)=\rho_i\,\theta_S'(q;S_i).\tag{47}$$

Multiplying (47) by $\frac{1}{4\pi}\theta_S(q;\alpha_3,\epsilon_3)$ and integrating over $q\in\mathbb{R}$ immediately gives $I_{SD}(3;q_{i+1},\eta)-I_{SD}(3;q_i,\eta)=\rho_i\,I_{SS}(3,S_i)$. Summing over $i=1,2,3$ telescopes the left-hand side to zero, proving $\sum_{i=1}^{3}\rho_i\,I_{SS}(3,S_i)=0$. By contrast, the extra boundary term $\frac{\pi}{2}\delta_{\epsilon_3,\epsilon_{S_i}}$ in $J_{SS}(3,S_i)$ comes from integrating $\theta_S(q;S_i)\theta_S'(q;3)$ by parts across the soliton zero-mode jump $\theta_S(0^+)-\theta_S(0^-)=-2\pi$, which has no counterpart in $\theta_D^{\mathrm{phys}}(0^\pm)=0$ and is *not* a coboundary on the KP vertices $\{q_1,q_2,q_3\}$. □

What Claim 6.1 implies for the quantum two-soliton S-matrix. In Phase 2e (`oneloop_phase.pdf`), we noted a $O(\hbar^0)$ convention choice for same-direction collisions ($\epsilon_1=\epsilon_2$) between the raw Faddeev–Korepin integral I_{12} (which satisfies $I_{12}(\alpha,\alpha)=0$, corresponding to bosonic normalisation $S_{\mathrm{same}}(\alpha,\alpha)=+1$) and the antisymmetrised integral $J_{12}=I_{12}+\frac{\pi}{2}\delta_{\epsilon_1,\epsilon_2}$ (which approaches $+\frac{\pi}{2}$ as $\alpha_1\to\alpha_2$, corresponding to fermionic/exclusion normalisation $S_{\mathrm{same}}\to$ i). Claim 6.1 provides a decisive algebraic consistency selection rule from integrable defect fusion:

Defect-fusion selection rule (c): *Only the raw Faddeev–Korepin one-loop phase I_{12} (together with the momentum-renormalisation term $-\Delta X_{12}\Delta P$) is an exact coboundary on the space of KP vertices, $I_{12}(3,S) = I_{SD}(3;p_b,+1) - I_{SD}(3;p_a,+1)$, and therefore satisfies one-loop defect fusion (44) and closes around every KP triangle (45).*

6.3 Small-amplitude limit, absorption lines, and stability/cusp jumps

We record the limiting regimes of $I_{SD}(\alpha,\epsilon;s,\eta)$ and $\delta^{(q_D)}_{D,\mathrm{renorm}}(\alpha,\epsilon;\eta)$ **(c)** `[oneloop_defect_phase_and_fusion.p`

1. **Small-amplitude limit $\alpha \to 0$:** Using $L(w\,\alpha/2) = -w\alpha(\ln(\alpha/2) + \ln|w| - 1) + O(\alpha^3 \ln\alpha)$ for $\phi_3 = \alpha$ and the Taylor expansion of $L(\tan(\phi_{1,2}/2)w)$ around $\phi_{1,2} = \pm 2\pi/3$, $I_{SD}(\alpha,\epsilon;s,\eta)$ vanishes as $O(\alpha\ln\alpha) + O(\alpha)$:

$$I_{SD}(\alpha,\epsilon;s,\eta) = \frac{\epsilon}{2\pi}\sum_{s'=1}^{2} c_{D,s'}\ \mathrm{Re}\Big[-2w_{s'}\alpha\big(\ln(\alpha/2)+\ln|w_{s'}|-1\big)+4w_{s'}\alpha\,L'\big(\sqrt{3}\,w_{s'}\big)\Big]+O(\alpha^3\ln\alpha), \tag{48}$$

   where $w_{s'} := (\tan(\psi_{s'}/2))^{-\epsilon}$ and $L'(z) = \frac{2\ln|z|}{1-z^2}$ (verified at $\alpha = 10^{-2}, 10^{-4}, 10^{-6}$ with error scaling as $1.57\times10^{-7}, 3.45\times10^{-13}, 5.33\times10^{-19}$).

2. **Soliton absorption lines $q_D \to p_a(\alpha,\epsilon)$ or $p_b(\alpha,\epsilon)$:** At an absorption threshold where the soliton chord shares a vertex with the defect ($z_1 \to 0$ or ∞), none of the arguments $\tan(\phi_r/2)(\tan(\psi_{s'}/2))^{-\epsilon}$ equals ±1 (since ϕ_r encodes the root differences of the soliton triple while $\psi_{s'}$ encodes the root differences of the defect triple, which only coincide if the two triples are identical). Consequently, the continuum integral $I_{SD}(\alpha,\epsilon;s,\eta)$ remains **smooth and finite** across the absorption line (e.g. at $s = 0.7$, $\epsilon = -1$, $\alpha_{\mathrm{abs}} \approx 0.659088$, $I_{SD}(\alpha_{\mathrm{abs}},-1;0.7,1) = 0.7225885601$), whereas the fixed-physical-momentum shift $\delta^{(q_D)}_{D,\mathrm{renorm}} \propto \ln|z_t|\,\Delta P_1$ diverges logarithmically as $\pm k^{-1}\Delta P_1(\alpha_{\mathrm{abs}},\epsilon)\ln|\alpha-\alpha_{\mathrm{abs}}|$.

3. **Jumps across the defect stability threshold $s = -\epsilon$ and the soliton cusp $\alpha = \pi/3$:** When a parameter crosses a locus where $\tan(\phi_r/2)(\tan(\psi_{s'}/2))^{-\epsilon} = \pm1$, $\mathrm{Re}\,L(z)$ has a finite slope while the integral representation (33) jumps by $\pm\pi$ as a pole $q = \pm\sin\psi_{s'}$ crosses $q = 0$ or merges with a soliton pole:

   - Across the defect stability threshold $s = -\epsilon$ ($\gamma = 2\pi/3$ for $\epsilon = +1$, where $\psi_1 = 0$, or $\gamma = \pi/3$ for $\epsilon = -1$, where $\psi_2 = -\pi$), $\tan(\psi_{s'}/2)^{-\epsilon}$ changes sign through 0 or ∞, and $I_{SD}(\alpha,\epsilon;s,\eta)$ jumps by $-\epsilon\eta\pi$ (verified at $s = -1 \pm 10^{-10}$ for $\epsilon = +1$: jump $-\pi$; and at $s = +1 \pm 10^{-10}$ for $\epsilon = -1$: jump $+\pi$).
   - Across the soliton cusp $\alpha = \pi/3$ ($\phi_1 = \pi$, $\tan(\phi_1/2) \to \pm\infty$), $I_{SD}(\alpha,\epsilon;s,\eta)$ jumps by $+\epsilon\eta\pi$ for $|s| > 1$ and is continuous (0 jump) for $|s| < 1$ (verified at $s = -1.5$: jump $+\pi$; at $s = 0.5$: jump 0).

7 Summary of claims, verification scripts, and failures recorded

Failures recorded during the investigation

In accordance with the project ground rules (`AGENTS.md`), we record all failed checks and algebraic errors encountered and corrected during this task:

1. **Equal weights $(+1,+1)$ instead of dipole weights $(-\eta,+\eta)$ in the 2-root decomposition of $\theta'_D(q;s,\eta)$.** When first testing the 2-root decomposition of $\theta'_D(q;s,\eta)$ at the angles $\psi_{1,2} = \gamma \pm \pi/3$, we assigned equal weights $(+1,+1)$ to $\Theta'(q;\psi_1)$ and $\Theta'(q;\psi_2)$, which produced $O(1)$ residuals (0.31). Carrying out the exact partial-fraction decomposition of $\theta'_D(q;s,\eta)$ showed that the two roots enter with opposite dipole weights $(c_{D,1},c_{D,2}) = (-\eta,+\eta)$ at $\psi_1 = 2\pi/3 - \gamma$ and $\psi_2 = -2\pi/3 - \gamma$, reducing the residual to 1.88×10^{-36}.

Claim / Formula	Label	Verification script	Precision / Result
3-soliton Hirota solution (11) on the CZ defect satisfies all 4 nonlinear sewing conditions at $x = 0$	(c)	`oneloop_defect_scattering_and_stability.py`	Max residual 1.20×10^{-34}
Exact time-dependent phonon eigenmodes on 1- and 2-soliton + defect backgrounds (Claim 2.1): $R(q) = 0$ and factorised $T_{\mathrm{tot}}(q)$ (14) for all $t \in \mathbb{R}$	(c)	`oneloop_defect_scattering_and_stability.py`	Max linearised sewing residual 3.94×10^{-34} ($N = 1$), 2.10×10^{-34} ($N = 2$)
KP-vertex/chord factorisation (18) and 2-root $(-\eta, +\eta)$ dipole decomposition of $\theta'_D(q; s, \eta)$ (23) (Claims 3.1–3.2)	(c)	`oneloop_defect_scattering_and_stability.py`	Max residual 6.48×10^{-36} across $\lvert s\rvert < 2$ and $\lvert s\rvert > 2$
Closed-form factorisation of the 4×4 localised-mode determinant (25), exact growing-mode eigenvalues (29), and Levinson jumps (30) (Claim 4.1)	(c)	`oneloop_defect_scattering_and_stability.py`	Det. residual 3.89×10^{-36}; root residual 1.62×10^{-35}; Levinson residual 6.12×10^{-34}
Equality $I_{SD} = J_{SD}^{\mathrm{phys}}$ (33), closed-form 3×2 dilogarithm formula (34), crossing symmetry (36), and FK α-gradient (Claim 5.1)	(c)	`oneloop_defect_phase_and_fusion.py`	Quadrature match 4.51×10^{-36}; crossing 2.26×10^{-36}; FK gradient 9.40×10^{-38}
Closed-form defect vacuum shifts $(\Delta E_D^{\mathrm{ren}}, \Delta P_D)$ (37)–(38) and exact one-loop (E, P) balance in soliton absorption (40) (Claim 5.2)	(c)	`oneloop_defect_phase_and_fusion.py`	Quadrature match 7.52×10^{-37}; balance across 18 absorption edges 3.20×10^{-36}
Exact one-loop defect fusion (44), KP-triangle closure for I_{SS} (45), and coboundary obstruction for J_{SS} (46) (Claim 6.1)	(c)	`oneloop_defect_phase_and_fusion.py`	Tested on 12 KP triangles (36 edges): I_{SD} fusion 4.51×10^{-36}, triangle closure 1.50×10^{-35}
Small-amplitude $O(\alpha \ln \alpha) + O(\alpha)$ law (48), finiteness at absorption lines, and $\pm\pi$ stability/cusp jumps	(c)	`oneloop_defect_phase_and_fusion.py`	Small-α residual 5.33×10^{-19} at $\alpha = 10^{-6}$; jump errors $< 10^{-7}$

Table 1: Summary of claims and verification checks in Phase 2g (Route E, step 2).

2. **Sign error in the symbolic 4×4 localised-mode determinant.** In our initial symbolic transcription of the 4×4 linearised CZ sewing matrix for localised modes, we flipped the sign of the term $\frac{\sqrt{3}}{6}\eta\sigma(\sigma^2-1)(\kappa_1-\kappa_2)$ in two rows, which produced an irreducible quintic polynomial in $S = \kappa_1 + \kappa_2$ giving $\gamma_{\text{grow}} = -0.026$ instead of $+0.02863$ at $\eta\sigma = -0.95$. Restoring the exact sign from row 2 and row 4 of the CZ sewing matrix caused the determinant to factor completely into $-\frac{S^2(S^2-1)}{24}P_-(S;\eta\sigma)P_+(S;\eta\sigma)$, matching the numerical 4×4 determinant to 3.89×10^{-36}.

3. **Spurious $1/\sqrt{3}$ factor in $\sqrt{1-3q_D^2}$ in Check 3 of `oneloop_defect_scattering_and_stability.py`.** When checking $\sin\psi_1 = q_+ - q_D$ for $q_D = -(\cos\gamma)/\sqrt{3}$, we initially wrote $\sqrt{1-3q_D^2} = (\sin\gamma)/\sqrt{3}$ instead of $\sin\gamma$, giving a mismatch of 0.207. Removing the spurious $1/\sqrt{3}$ reduced the residual to 1.88×10^{-36}.

4. **Branch cut in $I_{SD}(\alpha,-1;s,\eta)$ and $(\Delta E_D^{\text{ren}}, \Delta P_D)$ for $|s|>2$.** Evaluating $I_{SD}(\alpha,-1;s,\eta)$ by sending $\phi_r \mapsto \pi - \phi_r$ while keeping $\tan(\psi_{s'}/2)^{-1}$ unflipped shifted the integral by π for $s < -1$; using the exact crossing relation $\tan((\pi-\phi_r)/2) = \cot(\phi_r/2)$, which replaces $\tan(\psi_{s'}/2)^{-1}$ by $\tan(\psi_{s'}/2)^{\epsilon}$ and multiplies by ϵ, gave agreement to 4.51×10^{-36} for all $s \in \mathbb{R}$. Similarly, writing $\ln|\tan(\psi_{s'}/2)|$ instead of the analytic function $\frac{1}{2}\ln\frac{1-\cos\psi_{s'}}{1+\cos\psi_{s'}}$ in ΔP_D and ΔE_D^{ren} introduced a 0.2073 branch-cut error for $|s|>2$ ($\gamma = \pi + \mathrm{i}\beta$); using $\frac{1}{2}\ln\frac{1-\cos\psi_{s'}}{1+\cos\psi_{s'}}$ gave agreement to 7.52×10^{-37}.

5. **Incomplete ψ-derivative of $\Theta(q;\psi)$ and complex logarithm in tree-level fusion check.** In the first run of `oneloop_defect_phase_and_fusion.py`, an over-simplified expression for $\partial_\psi\Theta(q;\psi)$ gave a residual of 2.1 in the Faddeev–Korepin α-gradient check, and using $\ln(A_{12})$ instead of $\ln|A_{12}|$ in δ_0 raised a `TypeError` when $A_{12} < 0$. Furthermore, comparing $I_{SD}(3;q_{i+1},\eta) - I_{SD}(3;q_i,\eta)$ directly against $\rho_{\text{tree},i}I_{SS}(3,S_i)$ without accounting for the directional transmission factor $\rho_{\text{tree},i} = \epsilon_3\rho_i$ gave a mismatch for $\epsilon_3 = -1$. Fixing $\partial_\psi\Theta(q;\psi)$, using $\ln|A_{12}|$ and $\ln|z_t|$, and distinguishing $\rho_i = \eta\,\mathrm{sgn}(q_i - q_{i+1})$ from $\rho_{\text{tree},i} = \epsilon_3\rho_i$ reduced all residuals to $< 1.67 \times 10^{-33}$.

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